

Storage Reservoir

It is necessary to store water during excess flow periods (rainy season) so that same may be used during lean flow periods.

Low head plants require very big storage capacity of storage reservoir depends on the difference between runoff during high and lean flows, and detailed hydrological studies over long periods are necessary. (Indira Sagar dam has largest reservoir capacity in India on Narmada river.)

DAM

Function of dam in a hydroelectric project is to create an artificial head and storage. It is most expensive part of hydro project.

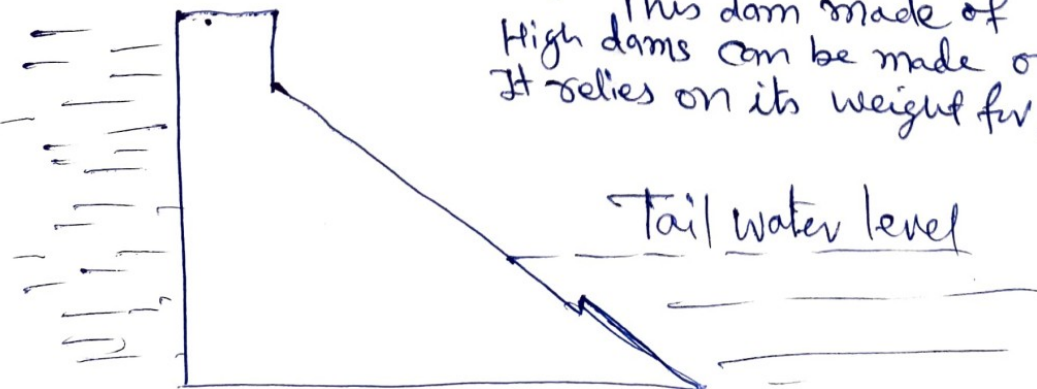
Broadly two types of dams

- (1) Masonary dams (2) Earth dams. (Hirakund dam in ~~India~~ India)

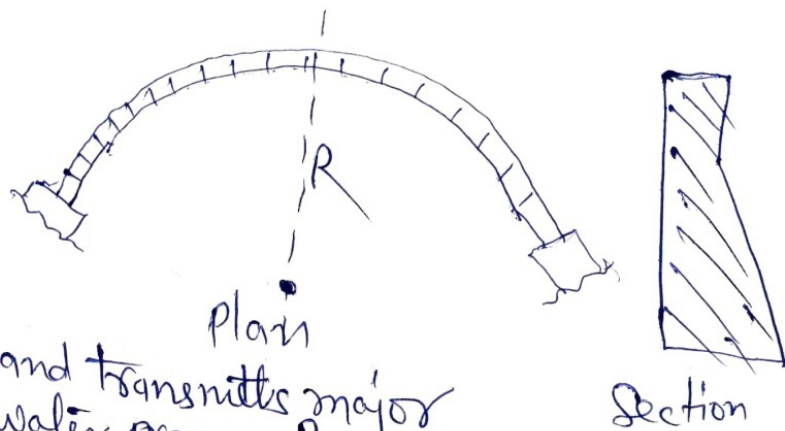
Masonry dams: Further classified as

- (i) Solid Gravity concrete dam:

This dam made of concrete. High dams can be made on rocky soil. It relies on its weight for stability of the dam.



(ii) Arch Dam:



It is curved and transmits major portion of its water pressure horizontally to the abutments by arch section. It transmits ~~load~~ load depends on the radius of curvature.

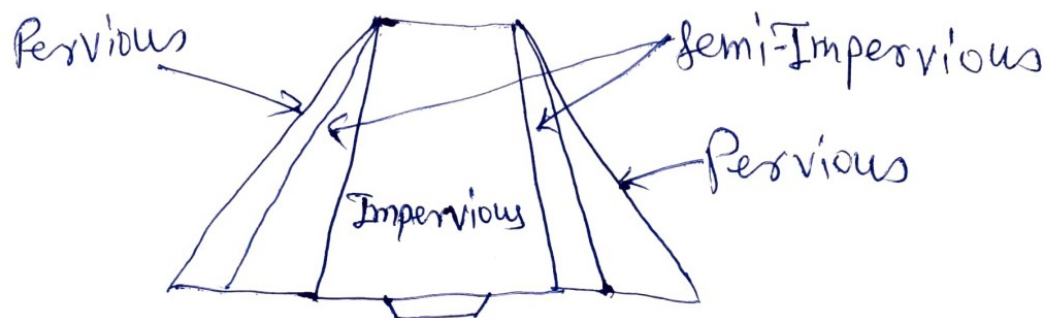
(iii) Buttress Dam: It has an inclined upstream face and buttress. The buttress transmits the water load and weight of the dam to the foundation. Such dam is more suitable for weak foundations and earthquake prone sites.

(2) Earth dam:—

An earth dam has a very wide base as compare to its height. It can be built on earthen or rocky foundations. An earth dam is cheaper than solid gravity dam.

Advantages:

1. They are suitable for relatively pervious foundation
2. They are cheaper than masonry dams.
3. They are in harmony with natural surroundings



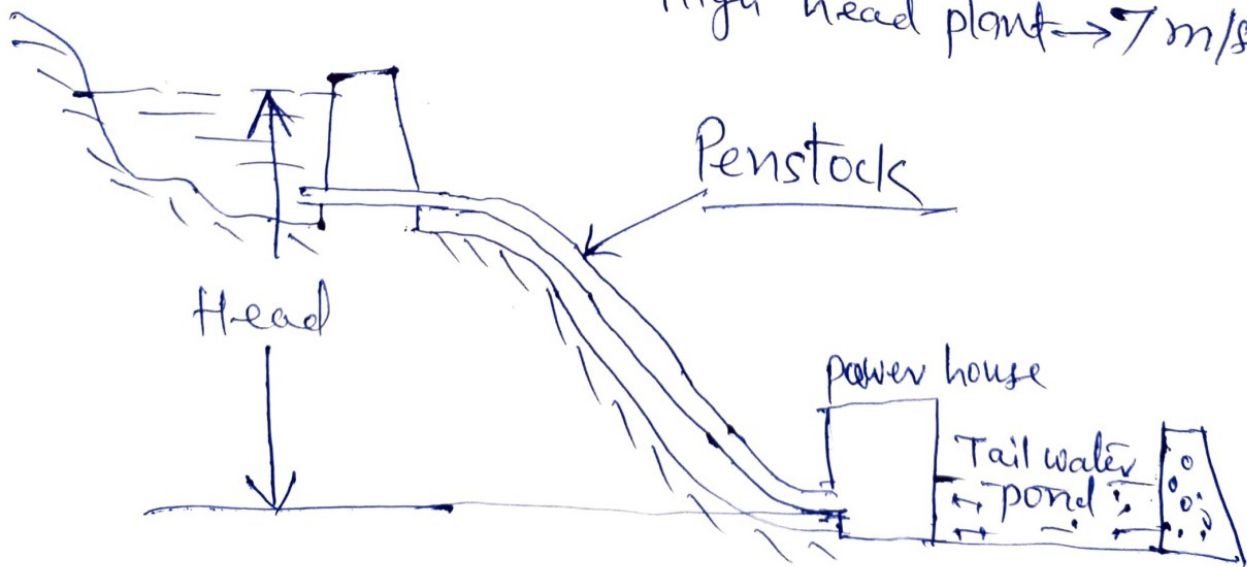
Disadvantage

- 1) Subject to erosion
- 2) not suitable for spillway, a supplementary spillway of adequate capacity is needed
- 3) The seepage of water is more as compare to that in masonry dam.

Penstock:— From intake and forbay water is taken to the turbine by a conduit system known as a penstock. Two types of penstock are used. (i) High pressure and (ii) Low pressure. High pressure consists of steel pipes which can take water under pressure. Low pressure type consists of a canal, a flume or a pipe. Penstock pipes are generally of steel for high and medium-head plants, they may be of concrete in the low head plants. Diameter and cross sectional area of a penstock pipe would be smaller if it were possible for the velocity of the water to be high.

Velocity of water →

- Low head → 2 m/sec.
- Medium head plant → 4 m/sec.
- High head plant → 7 m/sec.



Forebay: —

It is an enlarged body of water just above the intake ~~to the turbine~~. It may be enlarged section of the canal or a pond, to accommodate the required width of intake.

Intake: — The function of Intake in a hydro-electric project is to let the water into the conduit or penstock under controlled conditions. It is provided with trash racks, screens and booms to divert and prevent entry of debris and ice into the turbine. It is classified as High pressure intake and low pressure intake. High pressure one used in the case of big Storage reservoir and low pressure intake used for small ponds.

Surge tank: —

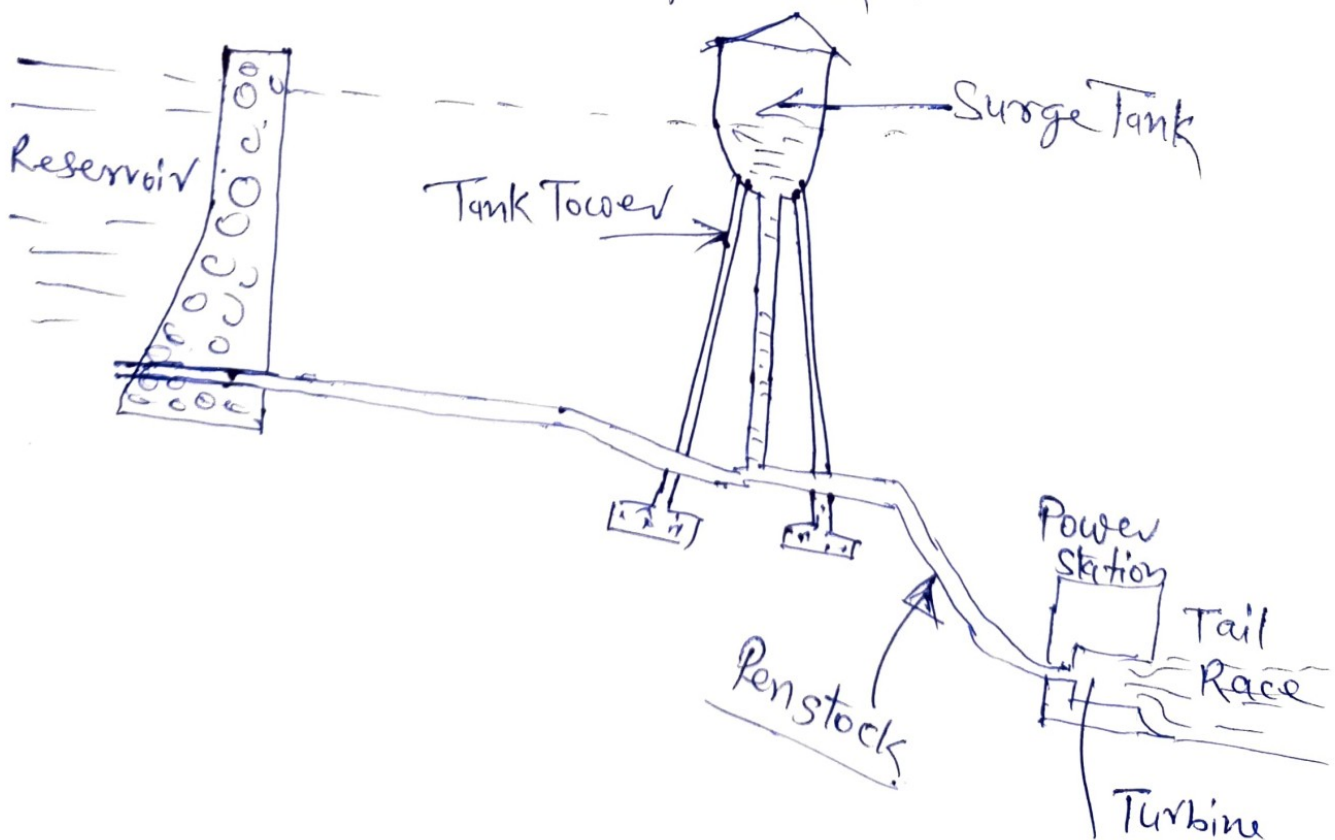
The load on generator keeps on fluctuating. Therefore, gates admitting water to the turbine are suddenly closed when load on generator suddenly reduced. There is a sudden change of pressure in the penstock pipes caused by sudden change in the rate of water flow. This sudden change of pressure above normal is known as "Water hammer". When load on generator increased suddenly, turbine gates suddenly open ~~back~~ to supply more water for increased load. Water has to rush through

the penstock, which causes Vacuum or a negative pressure in the water system.

The penstock must withstand the positive water hammer (when gates are closed) and Vacuum when gates are opened.

To provide better regulation of water pressure in the system, a surge tank is used. Surge tank is placed as close to the power station as possible.

When the gates are closed and the water flow into the turbine is suddenly reduced, water rises in the surge tank; and vice-versa. When load increases suddenly, surge tank provides additional water which increases flow of water in the penstock.



Spillway:-

Every dam is provided with an arrangement to discharge excess water during floods.

This may be a spillway or a by pass tunnel or conduit. Spillway should be so designed as to discharge the major flood waters without damage to the dam.

Spillways are classified as chute spillway, side channel spillway, shaft spillway, and Siphon spillway.

Hydraulic Turbines

Specific speed of Turbine (n_s): It is the speed at which a turbine would run if the runner were reduced to a size which would develop 1 metric h.p. under 1m head. This speed is proportional to square root of the horsepower and inversely proportional to the $\frac{5}{4}$ power of the head i.e.

$$n_s = \frac{n P^{1/2}}{h^{3/4}} \quad \text{RPM}$$

$n_s \Rightarrow$ specific rotational speed

$n \Rightarrow$ actual rotational speed

$P \Rightarrow$ Metric horse power

$h \Rightarrow$ Head (m)

In general the higher the specific speed for a given h.p. (o/p) and head (m) the lower is the cost of the installation as a whole.

A classification of turbines according to range of head and specific speed

Type of Turbine	Head	Specific speed (metric units)
Pelton	above 200m	10-50
Francis	30-200m	60-300
propeller	less than 30m	300-1000

PUMPED STORAGE PLANTS

8.34. GENERAL DESCRIPTION

A pumped storage plant is a special type of plant meant to supply peak loads. During peak load period, water is drawn from the head water pond through the penstock and generates power for supplying the peak load. During the off-peak period, the same water is pumped back from the tail water pond to the head water pond so that this water may be used to generate energy during the next peak load period. Thus, the same water is used again and again and extra water is needed only to take care of evaporation and seepage. Generally the pumping of water from the tail water pond to the head water pond is done at night when loads are low. The plant generates energy for supplying peak loads during day time. The off peak pumping maintains the firm capacity of the pumped storage plant. The reservoir capacity should be such that the plant can supply peak load for 4 to 11 hours. A general arrangement of pumped storage plant is shown in Fig. 8.20.

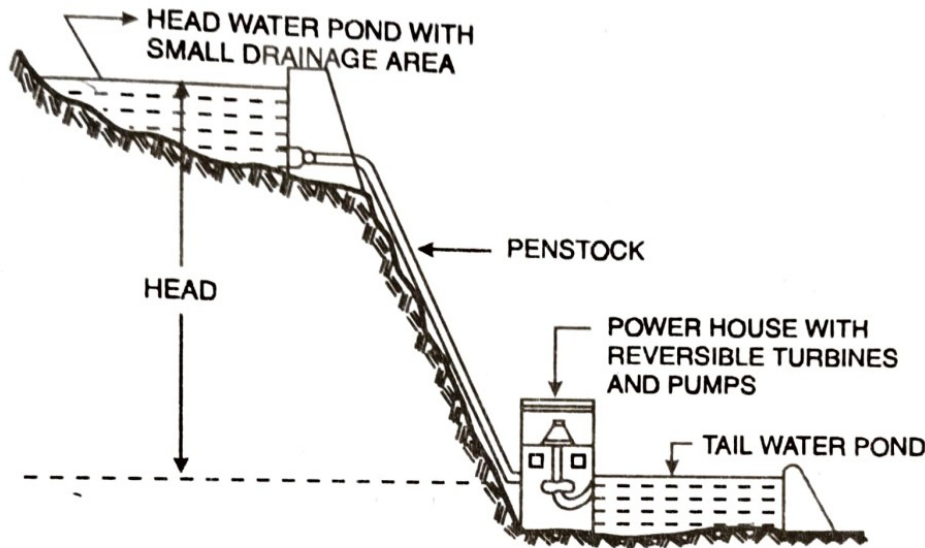


Fig. 8.20. Pumped storage hydro electric plant

The earlier pumped storage installations used a separate pump for pumping the water back into the head water pond. A recent development is a reversible turbine pump. During peak loads, the turbine drives the alternator and the plant generates electrical energy. During low loads, the alternator runs as a motor and drives the turbine which now works as a pump for pumping the water into the head water pond. This arrangement reduces the capital cost of the plant. The power for driving the motor is taken from the system.

8.35. PUMPED STORAGE PLANTS FOR SUPPLYING PEAK LOADS

Every pumped storage scheme requires dual conversion of energy, the efficiency being around 60 to 70 percent. The fact that the energy gained from a pumped storage plant is always less than the energy input, should not obscure the fact that this apparent loss to the system is negligible when compared with the substantial savings in fuel which are made when these plants are used in a mixed system. The most obvious economies are in the replacement of standby thermal plants and in the fact that the existing thermal plants can be run with fewer and less rapid load changes.

Modern generation systems are a mixture of hydro, steam and nuclear power stations so that energy can be generated most economically. If hydro contribution is small, the steam and nuclear stations have to be of sufficiently large capacity. The steam and nuclear plants operate more economically when run continuously at or near full load. At the time of low loads the excess energy generated by steam and nuclear plants can be used for driving the motor of the pumped storage scheme. The pumped storage plant would in turn supply peak loads. Thus the capacity of the steam and nuclear plants need be only for supplying the base loads and hence much less than what it

would be if they were to meet the whole demand. In addition the steam and nuclear plants can be worked at almost unity load factor which ensures their most economic operation.

8.36. ADVANTAGES

Pumped storage plants have very important advantages. Some of these are :

- (i) peak loads can be supplied at a lesser cost than that incurred when these loads are supplied by steam and nuclear plants.
- (ii) standby capacity is available on short notice. Power engineers in utilities having pumped storage installations have long realized the benefit of quick switching on and off capability of these installations. Pumped storage plants need a starting time of only 2 to 3 seconds and can be loaded fully in about 15 seconds. In the event of an outage on a unit, a pumped storage plant can be called upon to meet the generation deficiency (occurring due to the outage) thus ensuring reliable supply and avoiding the necessity of load shedding.
- (iii) the load factor of steam and nuclear plants is increased thus ensuring their efficient and economic operation.
- (iv) since the base loads plants need not be used to supply peak loads, the forced and maintenance outages of these plants are likely to be reduced.
- (v) in the event of an extra demand coming up suddenly on the system, these plants can be immediately switched on to meet this extra demand.
- (vi) because of their ability to take up loads in a very small time, the use of pumped storage plants reduces the spinning reserve requirement of the system.
- (vii) they can be used for load frequency control.

8.37. TYPES OF PUMPED STORAGE INSTALLATIONS

Pumped storage plants offer many benefits. However their feasibility and economics should be critically evaluated at the planning stage. During the power system planning studies, three types of pumped storage installation can be considered :

- (i) high head, daily/weekly cycle, peaking plants close to the load centres.
- (ii) low/medium head, seasonal cycle plants, far-away from load centres (peaking or non-peaking type).
- (iii) medium head, daily/weekly cycle peaking plants at an existing cascade of reservoirs.

The expected benefits from these three schemes are different. The daily/weekly cycle plants provide an efficient and economical way to provide the peak power to the system. On the other hand the low/medium head seasonal plants are intended to increase the regulated flow to down stream plants, converting the secondary energy available during flood periods into useful firm energy generated in critical or drought periods. Most of the countries in the world have sufficient potential of one or more of the above types of pumped storage schemes.

8.38. DUTIES AND CHARACTERISTICS

Depending on the advantage to be realized, the duties of a pumped storage plant can be classified into static duties (*viz.* the power duty and energy transfer duty) and dynamic duties (*viz.* spinning reserve duty, load frequency control duty and pick up duty).

The characteristics of these plants which affect the above duties are :

- (i) static characteristics *i.e.*, rated generating capacity W_g (MW), rated pumping capacity W_p (MW), energy storage capability E (MW hrs.), cycle efficiency (percent), number of equivalent generating hours at full generating capacity. *i.e.*, $h = E/W_g$ (hours) and restoring ratio *i.e.*, $r = W_p/W_g$ (p.u.).
- (ii) dynamic characteristics *i.e.*, time of start up (seconds), loading rate of units (MW per second).